

BXP Evidence Portfolio

A Structured Overview for External Reviewers

Project: Breathe Exposure Protocol (BXP)

Creator: Elvarin

Category: Independent research in open data standards for public health

Repository: <https://github.com/bxpprotocol/bxp-spec>

Specification DOI: <https://doi.org/10.5281/zenodo.18906812>

Implementation DOI: <https://doi.org/10.5281/zenodo.18907003>

Project Overview

BXP (Breathe Exposure Protocol) is an independently developed open data standard for atmospheric exposure information — a universal file format and REST API specification that allows any air quality measurement, from any source, anywhere in the world, to be represented, stored, and exchanged in a consistent, machine-readable format.

The project emerged from an investigation into why air quality data — despite the proliferation of affordable sensors — remains fragmented, incompatible, and therefore underused. BXP proposes and partially demonstrates that this fragmentation can be eliminated through an open protocol standard: a common language for breath exposure data that no company owns and anyone can implement.

Original Problem

Air pollution kills approximately 7 million people per year (WHO, 2021). The sensors to measure it exist and are affordable. The barrier is not hardware — it is data fragmentation. Every sensor manufacturer, monitoring agency, and research network uses incompatible data formats. A sensor in Accra cannot feed a dashboard in Nairobi. A citizen reading in Delhi cannot contribute to a government pollution map. The data exists, but it cannot be used together.

Original Insight

The insight that led to BXP was recognising this as an interoperability problem rather than a measurement problem. The solution was not to build a better sensor or a better application — it was to define a common data language, in the same way that HTTP solved the web's interoperability problem and MP3 solved audio fragmentation.

What I Personally Did

Everything in this repository was independently created by Elvarin. This includes:

- * Identifying and documenting the air quality data interoperability problem
- * Designing the protocol architecture and data model from first principles
- * Writing the formal BXP v2.0 specification (1,300 lines)
- * Building the reference server in Python (FastAPI, 1,877 lines)
- * Building the Python SDK (895 lines)
- * Building the TypeScript SDK
- * Building the CLI tool (740+ lines)
- * Writing the API documentation, developer guide, and architecture documentation
- * Publishing the specification and implementation to Zenodo with permanent DOIs
- * Conducting the systematic 50-item gap analysis between v1.0 and v2.0
- * Creating the sample dataset representing 10 global cities

Research Conducted

- * Investigation of existing air quality data standards and their limitations (US AQI, European CAQI, OpenAQ, OGC SensorThings, HL7 FHIR)
- * Study of WHO Air Quality Guidelines 2021 for threshold values
- * Study of DALY (Disability-Adjusted Life Year) burden data for HRI weighting
- * Analysis of interoperability approaches in other domains (HTTP, PDF, MP3, DICOM)
- * Investigation of privacy frameworks for personal health data (GDPR, CCPA, POPIA)
- * Study of cryptographic techniques for data integrity (SHA-256, Ed25519)

Technical Work

Component	Technology	Lines	Status
Protocol specification	Markdown	~1,300	Complete
Reference server	Python/FastAPI/SQLite	~1,877	Complete
Python SDK	Python	~895	Complete
TypeScript SDK	TypeScript	~400	Complete
CLI tool	Python	~740+	Complete
MQTT bridge	Python	~200+	Complete
Docker deployment	Docker	-	Complete
Sample dataset	JSON	250 lines	Complete
API documentation	Markdown	~200	Complete
Developer guide	Markdown	~200	Complete

Protocol Development

The protocol design work included:

- * Defining a canonical agent schema for 31 atmospheric substances across 7 categories
- * Designing the BXP_HRI composite health risk index with WHO-derived weights
- * Specifying a binary file format with 32-byte fixed header (magic number, version, flags, checksums)
- * Designing a five-stage data pipeline (LOCATE → DETECT → INTERPRET → PROTECT → REPORT)
- * Specifying 20+ REST API endpoints with full request/response schemas
- * Designing a privacy framework with geohash floors, k-anonymisation, and cryptographic deletion

- * Specifying a community reporting layer with 12 standardised event tags
 - * Designing a governance model with RFC process and versioning policy
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Prototype / Implementation

The reference server is a functional prototype that:

- * Accepts readings from any source via HTTP
 - * Validates data against the BXP specification
 - * Computes BXP_HRI health risk scores
 - * Stores readings in SQLite with SHA-256 payload hashes
 - * Enforces privacy controls (geohash floor, $k \geq 5$ anonymity, cryptographic deletion)
 - * Serves an interactive web dashboard with a world map and time-series charts
 - * Can be started with four Python packages and a single command
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Documentation

Seven documentation artefacts were created:

1. SPEC.md — formal protocol specification
 2. docs/api_documentation.md — API reference
 3. docs/developer_guide.md — developer quick-start
 4. docs/protocol_overview.md — protocol overview
 5. CHANGELOG.md — development history
 6. CONTRIBUTING.md — contribution guide
 7. This evidence package (9 documents)
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Evidence of Iteration

The development history demonstrates clear iteration:

- * v1.0 (February 15, 2026): First prototype — portable containers with SHA-256 verification, offline-first design
- * Gap analysis: 50-item systematic audit identifying missing protocol components
- * v2.0 (March 2026): Complete specification — full agent schema, REST API, privacy framework, HRI with modifiers, governance model
- * Reference server v2.1: All critical gaps from the audit addressed — persistence, auth, privacy, rate limiting, dashboard, community reports, etc.
- * SDK and CLI expansion: Async client, offline queue, TypeScript SDK, batch operations added

This is documented in CHANGELOG.md and the development notes.

Current Status

BXP v2.0 is a complete open protocol specification with a working reference implementation, Python SDK, TypeScript SDK, CLI tool, and ecosystem tooling. It has been published to Zenodo

with permanent DOIs. The GitHub repository is public.

Honest assessment of maturity: BXP is a research prototype and protocol proposal. The binary file format is specified but not yet implemented in software. The federated network is designed but not yet built. No third-party has independently implemented the protocol. These are the natural limitations of independent research conducted by one person.

Future Direction

- * Binary format implementation
 - * Federated node synchronisation
 - * Arduino/ESP32 SDKs for hardware sensors
 - * BXP-STREAM real-time data extension
 - * External review of BXP_HRI by atmospheric scientists/epidemiologists
 - * Engagement with sensor manufacturers or monitoring organisations
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Evidence Table

Evidence Item	What It Demonstrates	Location
SPEC.md (1,300 lines)	Protocol design and systems thinking	GitHub root
Reference server (reference-server/)	Technical implementation ability	GitHub
Python SDK (sdk/python/bxp_sdk.py)	Software development	GitHub
TypeScript SDK	Cross-platform development	sdk/typescript/
CLI tool (cli/bxp_cli.py)	Tooling and developer experience des	GitHub
CHANGELOG.md	Iterative development and improvemen	GitHub root
Sample dataset (10 cities)	Global scope of the protocol	datasets/
Development gap analysis (50 items)	Systematic thinking and self-critiqu	attached_assets/
MQTT bridge	IoT integration thinking	integrations/
Docker deployment	Production-ready thinking	Root
Zenodo DOI (spec)	Independent publication	doi.org/10.5281/zenodo.18906812
Zenodo DOI (implementation)	Independent publication	doi.org/10.5281/zenodo.18907003
GitHub repository	Public, reproducible, open-source	github.com/bxpprotocol/bxp-spec
This evidence package	Communication and research documenta	BXP_Evidence_Package/

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